

Module 3: Data Wrangling with Power Query

What is Power Query?

Power Query is an Excel data transformation tool used to import, clean, transform, and prepare data before analysis.

Imagine you receive messy data from different sources every day. Instead of cleaning it manually every time, Power Query lets you create a cleaning process once, and then reuse it with a single refresh.

Example

You receive a monthly sales file with:

- Blank rows
- Wrong data types
- Extra spaces
- Duplicate records

Instead of cleaning it every month manually, Power Query performs all those steps automatically after clicking **Refresh**.

Interview Point

Power Query is widely used in Excel and Power BI because it automates data preparation and reduces manual work.

One-Line Revision

Power Query automates data cleaning and transformation.

1. Importing Data

Importing Data means bringing data into Power Query from different sources for cleaning and analysis.

Before analyzing data, Excel needs to know where the data is coming from.

Common Data Sources

- Excel Workbook
- CSV File
- Text File
- Database
- Folder
- Web
- SharePoint

Interview Point

Power Query can connect to multiple data sources without manually copying and pasting data.

One-Line Revision

Importing loads external data into Power Query.

2. Cleaning Raw Data

A. Remove Nulls

Removes rows or values that contain missing (null) data.

Sometimes datasets have empty cells.

Removing or handling these missing values improves data quality.



Example

Name	Salary
------	--------

Ali	50000
-----	-------

Sara	<i>(Null)</i>
------	---------------

Ahmed	60000
-------	-------

After removing nulls:

Name	Salary
------	--------

Ali	50000
-----	-------

Name Salary

Ahmed 60000

Interview Point

Null values should be handled before analysis because they can affect calculations and visualizations.

One-Line Revision

Remove Nulls eliminates missing values.

B. Trim & Clean

Trim removes extra spaces, while Clean removes non-printable or hidden characters.

Imported data often contains invisible characters or unnecessary spaces that can cause matching errors.

Example

" Alisha Khan "

becomes

"Alisha Khan"

Interview Point

Use **Trim** for spaces and **Clean** for hidden characters.

One-Line Revision

Trim and Clean make text consistent.

C. Replace Errors

Replaces formula or import errors with another value.

Instead of displaying errors, Power Query lets you replace them with meaningful values like 0 or "Unknown".

Example

Error → 0

Interview Point

Replacing errors prevents calculations from breaking.

One-Line Revision

Replace Errors fixes invalid values.

D. Fixing Data Types

Assigns the correct data type to each column.

Excel should know whether a value is a number, date, text, currency, or percentage.

Example

Before	After
--------	-------

"2500" (Text)	2500 (Number)
---------------	---------------

Interview Point

Incorrect data types are one of the most common causes of analysis errors.

One-Line Revision

Correct data types ensure accurate analysis.

E. Fill Down

Copies the value from the cell above into blank cells below.

If a category name is written once and the following rows are blank, Fill Down automatically repeats the category.

Example

Category

Electronics

(Blank)

Category

(Blank)

After Fill Down:

Category

Electronics

Electronics

Electronics

Interview Point

Fill Down is commonly used when cleaning exported reports.

One-Line Revision

Fill Down fills blank cells using the value above.

3. Transforming Data

A. Split Columns

Separates one column into multiple columns based on a delimiter or position.

Power Query divides combined information into separate parts.

Example

Ali Khan



First Name Last Name

Ali Khan

One-Line Revision

Split divides one column into many.

B. Rename Columns

Changes column names to meaningful titles.

Instead of generic names like Column1, give columns clear business names.

Example

Column1 → Customer Name

One-Line Revision

Rename improves readability.

C. Extract

Extract retrieves a specific part of a text value.

Sometimes you only need part of the information.

Example

Email:

alisha@gmail.com

Extract:

gmail.com

One-Line Revision

Extract keeps only the required portion.

D. Transpose

Converts rows into columns and columns into rows.

It rotates the table layout.

Example

Before:

Name

Ali

Sara

After:

| Ali | Sara |

One-Line Revision

Transpose swaps rows and columns.

E. Pivot

Converts unique row values into separate columns.

Instead of repeating categories, Pivot creates a summarized table.

Example

Before:

Month Sales

Jan 100

Feb 150

After:

Jan Feb

Jan Feb

100 150

Interview Point

Pivot restructures data for easier reporting.

One-Line Revision

Pivot turns rows into columns.

F. Unpivot

Converts multiple columns into rows.

Unpivot does the opposite of Pivot and creates a "long" format dataset, which is often better for analysis and visualization.

Example

Before:

Jan Feb Mar

Jan Feb Mar

100 150 200

After:

Month Sales

Jan 100

Feb 150

Mar 200

Interview Point

Power BI and many analytics tools work better with unpivoted (long-format) data.

One-Line Revision

Unpivot converts columns into rows.

4. Adding Columns

A. Custom Column

Creates a new column using your own formula or logic.

You can build new information from existing columns.

Example

Profit = Sales - Cost

Interview Point

Custom Columns allow business-specific calculations during data preparation.

One-Line Revision

Custom Columns create calculated values.

B. Index Column

Adds sequential numbers to each row.

It gives every record a unique serial number.

Example

Index Customer

- 1 Ali
- 2 Sara
- 3 Ahmed

Interview Point

Useful for maintaining row order or creating unique identifiers.

One-Line Revision

Index Columns assign row numbers.

5. Merge & Append Queries

A. Merge Queries

Merge combines two or more tables using a common column (key).

It's like joining customer information with their order details using **Customer ID**.

Example

Customers Table

Customer ID Name

101	Ali
-----	-----

Orders Table

Customer ID Product

101	Laptop
-----	--------

Merged Result:

Customer ID Name Product

101	Ali	Laptop
-----	-----	--------

Interview Point

Merge is similar to **JOIN** in SQL.

One-Line Revision

Merge joins tables using a common key.

B. Append Queries

Append adds one table below another when both have the same columns.

Instead of joining side by side, Append stacks data vertically.

Example

January Sales



February Sales



March Sales



Combined Sales Table

Interview Point

Append is useful when combining monthly or yearly datasets.

One-Line Revision

Append stacks tables vertically.

6. Query Steps (Applied Steps)

Applied Steps record every transformation performed in Power Query.

Think of Applied Steps as Power Query's memory. Every action you perform is saved in order, allowing you to edit, remove, or replay those steps whenever new data arrives.

Example

Applied Steps might look like:

1. Imported Data

2. Changed Data Type
3. Removed Nulls
4. Trimmed Text
5. Split Column
6. Renamed Columns

When a new file is loaded, Power Query automatically repeats all these steps after clicking **Refresh**.

Interview Point

One of Power Query's biggest strengths is that every transformation is **repeatable and automated**.

One-Line Revision

Applied Steps store every transformation in sequence.